

BSSEC01/1577/2022

Project Proposal: AI-powered Document Search and Analysis Platform for SMEs in Kenya

Topic

Design and Implementation of an AI-powered Document Search and Analysis Platform for Kenyan SMEs

Background Work

Small and medium enterprises (SMEs) in Kenya generate large volumes of documents, including invoices, sales receipts, contracts, reports, and customer records. While these documents contain valuable information for business decision-making, most SMEs lack the tools and expertise to analyze them effectively.

Traditional systems only allow basic storage or keyword search, which limits the ability to extract meaningful insights. With advances in semantic search (using embeddings) and document analytics (e.g., trend detection, clustering), SMEs can now leverage AI to not only find information faster but also recognize business patterns hidden in their documents.

This project proposes a platform that enables SMEs to upload their documents, search semantically, and gain insights such as customer behavior trends, recurring financial patterns, or frequent issues raised in reports.

Specific Objectives

1. **To design and implement a semantic document search system** that enables SMEs to retrieve business information efficiently based on meaning rather than exact

keywords.

2. **To integrate analytical features** such as trend detection and clustering of related documents for pattern recognition.
 3. **To develop a decision-support dashboard** that visualizes insights from SME documents, helping business owners make informed decisions.
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Justification

SMEs in Kenya face challenges in managing and analyzing business data due to limited resources and lack of digital tools. As a result, valuable insights from documents such as sales receipts, supplier contracts, and customer feedback are often lost.

By combining semantic search with document analysis, this system will:

- Improve decision-making by revealing hidden patterns in business documents.
- Help SMEs recognize opportunities and risks (e.g., detecting rising customer complaints or seasonal sales trends).
- Provide a cost-effective AI solution tailored to the Kenyan SME ecosystem, where large enterprise systems are often too expensive or complex.

This project demonstrates how modern AI (embeddings, vector databases, text analytics) can be applied to real-world business problems, especially SMEs.